a FlexLifeFi K–12 program

GRADE 8 · LESSON PLANS

Ten Lessons, Ready to Teach

Standards, materials, procedure, differentiation — pick up Monday.

Year theme: Measure, Improve, Operate Smarter

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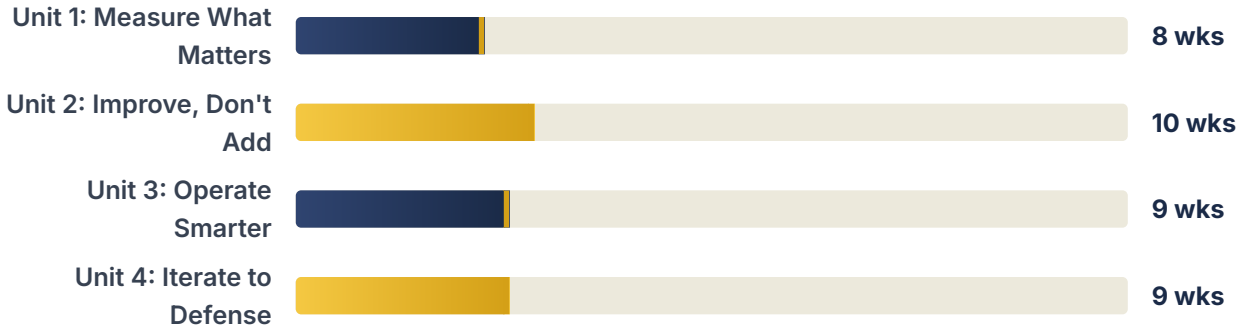
YEAR-AT-A-GLANCE

Four Units. Thirty-Six Weeks. One Capstone.

The year is built as four units, each ending with a public-facing artifact that feeds the rolling Capstone. The bars show how the weeks are spent — roughly equal, with a slight lean toward the math-heavy unit because per-customer math is the load-bearing skill.

WEEKS PER UNIT · GRADE 8



Each unit closes with a checkpoint artifact. Final defense in Week 36.

36 WEEKS

4 UNITS

10 LESSONS

8 CASH VIDEOS

GRADE 8 · LESSON PLANS

Ten Lessons, Ready to Teach

A little less wrong, every week.

Ten fully fleshed lessons spanning the year. Each lesson is built to pass the picks-up-Monday test: a teacher with no prior prep should be able to read this Sunday night and teach it Monday morning. Standards are cited at the benchmark level. Differentiation notes are specific and actionable. Vocabulary is included for each lesson. Time blocks default to 50 minutes (single middle-school period) unless noted; longer blocks are flagged.

Lessons are sampled across the year so the full arc is visible:

1. (Week 1) Cash V1 launch + Day 1 Improvement Bet
2. (Week 4) SCAMPER and How Might We — Structured Ideation
3. (Week 7) Value Levers — Reading a Business Case
4. (Week 8) Ideation Pack — Checkpoint 1 Studio Day
5. (Week 12) Linear-Equation Break-Even Modeling
6. (Week 16) Sensitivity — What If My Numbers Are Wrong?
7. (Week 22) Designing the 3-Day Mini A/B Test
8. (Week 26) Reading the Test — Bias, Lift, and What It Cannot Tell You
9. (Week 31) Variance — When Plan and Reality Disagree
10. (Week 36) Final Capstone Defense — Final PDSA Pack

Lesson 1 — Day 1 Improvement Bet

"Tell me what they actually said."

— CASH

Grade band: 6-8 **Time:** 50 minutes **Unit:** 1 — From idea to test **Topic area:** Continuous improvement · entrepreneurship launch

Objectives

By the end of this lesson, students will be able to:

- Write a "Day 1 Improvement Bet" — a single sentence naming one thing about their hypothetical Capstone venture they think would improve if tested.
- Distinguish between opinion (something I believe) and bet (something I'd run a small test on, with a metric).

Standards alignment

- **Jump$tart Standard 6 (Financial Decision Making), Benchmark 8-2:** Identify ways that consumers may avoid making poor decisions. (Applied to entrepreneurs running iterative tests rather than guessing.)
- **CCSS-ELA W.8.1:** Write arguments to support claims with clear reasons.
- **CASEL Self-Management:** Goal-setting and self-monitoring.

Materials

- Cash Video #1 ("From Idea to Test") — 2:50 (or printed transcript).
- One Capstone Journal per student.
- Two index cards per student.
- Wall poster: "OPINION → BET → TEST → STUDY → ACT" (the year's spine).
- A list of 8 sample statements, half opinions, half bets. Examples: "Kids would pay more for a colorful version" (opinion); "If I doubled the bottle size, I think 3 of 5 of my regulars would still buy at $5" (bet); "Marketing on TikTok is better than posters" (opinion); "If I posted at 4pm vs. 8pm, the 4pm post would get more views" (bet).

Vocabulary

- **Opinion** — something you believe without a way to check it.
- **Bet** — a sentence that names a change, a metric, and a guess about which way the metric will move.
- **PDSA** — Plan, Do, Study, Act. A four-step cycle for running small experiments to make a business better.

Procedure

1. Engage — Cash Video + One Question (~7 min)

Play Cash Video #1. As students watch, post the prompt on the board:

> *"What's one thing you'd change about something you do at home or school — and how would you know whether the change worked?"*

Pair-share for 2 minutes. Cold-call two students. (You're surfacing the muscle the year will train onto a venture: change something, measure something, learn something.)

2. Explore — Opinion vs. Bet Continuum (~12 min)

Distribute one card to each student. Read the rule:

> *"I'll read 8 statements. After each one, walk to the spot on the line that matches: pure opinion left, runnable bet right."*

Read each statement, pause, allow movement. After 3–4 statements, pick two students at different points and ask: "What made you pick that spot?" Don't correct — let students hear each other reasoning. Continue through all 8.

After the last one, ask the closing question: "What's the difference between an opinion and a bet in one sentence?" Two volunteers; sharpen on the board.

3. Explain — The Year's Spine (~8 min)

Define the three vocabulary words. Add to Vocabulary Spine.

State the year's first norm: "In this room, when you say something will improve a venture, you'll be asked: what's your bet, and what would tell you you're wrong? You'll ask each other this. I'll ask. By December, you'll be good at answering."

Walk through PDSA on the wall poster. Frame the year: students will run at least one full PDSA cycle on their own venture, and the Final Capstone Defense is a defense of what they learned from running it.

Cross-grade hook: name explicitly — "Last year (Grade 7), you learned who your customer is and how much each customer contributes after costs. This year, you'll learn how to run tests on top of that. We're not starting from scratch."

4. Apply — Write the Day 1 Improvement Bet (~15 min)

In the Capstone Journal, students write three things:

1. The kind of venture they're imagining (one sentence).
2. **The Day 1 Improvement Bet.** Format: "If I changed [X] in my venture, I think [metric] would move by [direction] because [reason]."
3. One thing they're least sure about in their bet.

Confer with 5 students while others write — read their sentence, ask one sharpening question.

5. Close — Pin and Predict (~8 min)

Each student writes the Day 1 Improvement Bet on an index card with their name. Collect them. Tell them you'll hand them back in Week 18 (after they've run their first PDSA cycle) so they can see how the bet held up under data. Project the exit prompt:

> *"Predict: in Week 18, will the data say you were right (1), partly right (2), wrong (3), or that the metric moved for a reason you didn't predict (4)? Write your number on the back of the card."*

Collect cards. End class.

Differentiation

- **ELL:** Provide the sentence frame on a sticky note: "If I changed **_ in my venture, I think would move by __** because ____." Allow native-language brainstorming with a peer; require English on the index card. Pre-teach opinion, bet, metric — Spanish cognates: opinión, apuesta, métrica.
- **IEP/504:** Allow a verbal version recorded on phone (with student transcribing later). Provide a printed list of the 8 statements during the continuum activity. Extended writing time on the bet — okay to finish in the next session.
- **Advanced learners:** Push for two metrics, not one. Ask: "Which metric would matter more if both moved? Why?" Write a second sentence: what would have to be true for the bet to be wrong even if the metric moved up.

Assessment

Formative: Watch the continuum activity for which students place statements correctly. Confer notes during writing. **Summative:** Day 1 Improvement Bet is filed; will be compared to Week 18 PDSA cycle for the Capstone reflection prompt.

Closure

Same as #5 above.

Homework / Extension

Read the page in the Student Activity Book titled Three Operate-Smarter Stories — three short founder accounts of running tests — and circle a sentence in any one where the founder names a metric.

Teacher notes

- The continuum is the engine. If the room can't accommodate movement, project the line and use sticky notes. Don't skip it.
- Some students will struggle to name a Capstone topic on Day 1 — that's fine. They'll lock the topic in Week 2. Many continue their Grade 7 venture.
- A student says "I don't have a venture." Coach: "What's something you ran or watched run last year? What's a piece of it you'd change?" Reach back into Grade 7 if helpful.
- Sensitivity flag: if a student names a venture involving family money, redirect to a hypothetical version. Curriculum norm: no probing of family finances.

Lesson 2 — SCAMPER and How Might We

Grade band: 6-8 **Time:** 60 minutes (block, or 2 × 30-minute periods) **Unit:** 1 — From idea to test **Topic area:** Structured ideation · divergent + convergent thinking

Objectives

By the end of this lesson, students will be able to:

- Apply SCAMPER (Substitute, Combine, Adapt, Modify, Put to other use, Eliminate, Reverse) to a sample problem and generate at least 7 distinct ideas.
- Convert one of those ideas into a "How Might We" (HMW) problem statement that names the user, the friction, and the desired outcome.

Standards alignment

- **NGSS-ETS MS-ETS1-1:** Define the criteria and constraints of a design problem with sufficient precision.
- **CCSS-ELA W.8.4:** Produce clear and coherent writing in which the development is appropriate to task.
- **CASEL Responsible Decision-Making:** Generate alternatives.

Materials

- SCAMPER reference sheet (provided), one per pair.
- HMW sentence-frame card (provided).
- Sample problem prompt: "Middle-school cafeteria lines are too long; some kids miss eating before class."
- Capstone Journal.
- Wall chart with 7 columns labeled S/C/A/M/P/E/R.

Vocabulary

- **SCAMPER** — a 7-prompt ideation tool: Substitute, Combine, Adapt, Modify, Put to other use, Eliminate, Reverse.
- **How Might We (HMW)** — a problem-statement format that turns a frustration into an open invitation: "How might we [do X] for [user] so that [outcome]?"
- **Divergent thinking** — generating many options.
- **Convergent thinking** — narrowing to the best option.

Procedure

1. Engage — The Cafeteria Problem (~8 min)

State the prompt aloud: "Middle-school cafeteria lines are too long; some kids miss eating before class."

Pair-share for 90 seconds: name three ideas to fix this. Cold-call three pairs; write ideas on the board. Resist evaluating. Note out loud: "Most of you went to the same 2–3 places. That's normal. We're going to push ourselves to go further."

2. Explain — SCAMPER (~10 min)

Walk through SCAMPER on the wall chart. Use the cafeteria problem to model one idea per prompt:

- **S — Substitute:** What could we substitute? "Substitute the line with assigned pickup windows."
- **C — Combine:** What could we combine? "Combine snack-grabbing with locker time."
- **A — Adapt:** What could we adapt from elsewhere? "Adapt a coffee-shop mobile order system."
- **M — Modify (or magnify/minify):** "Modify the menu to a smaller, faster set of choices."
- **P — Put to other use:** "Put the empty time before class to use — eat in homeroom."
- **E — Eliminate:** "Eliminate cashier checkout — pre-purchased meal cards."
- **R — Reverse (or rearrange):** "Reverse the flow — staff brings food to homerooms."

Mention: at least 4 of these ideas exist in real schools. Naming the technique matters — the technique is what produces options on purpose.

3. Explore — Pair SCAMPER on Your Own Capstone (~20 min)

In pairs, students apply SCAMPER to their own Capstone problem. They post 7 sticky notes on the wall chart (one per SCAMPER letter), each with one idea. Walk and confer; push pairs who post safe ideas to push further on at least one prompt.

After 12 minutes, partners do a fast-share with another pair: "Here are our 7. Which two surprised us?"

4. Explain — How Might We (~7 min)

Project the HMW frame: "How might we [do X] for [user] so that [outcome]?"

Walk through three examples drawn from the cafeteria problem:

- "How might we shorten lunch wait time for 7th-grade students so that they finish eating before class?"
- "How might we surface menu choices earlier for kids in PE so that they pre-decide?"
- "How might we reduce cashier delay for everyone so that lines move twice as fast?"

Note: a good HMW is specific enough to act on, open enough to allow many solutions.

5. Apply — Write 3 HMWs for Your Capstone (~12 min)

Students write 3 HMW statements for their Capstone, each in the full format. Confer with as many as possible. Common stuck point: "How might we make my product better for everyone so it's amazing" — too broad. Coach: name the user, name the friction, name the outcome.

6. Close — Vote and Park (~3 min)

Pair-share: each student reads their 3 HMWs to a partner. Partner picks the most "testable" one and explains why. Park HMWs in the Capstone Journal — they'll feed Lesson 4 (Ideation Pack Studio).

Differentiation

- **ELL:** Provide a SCAMPER reference sheet with cognate translations (Spanish: sustituir, combinar, adaptar, modificar, eliminar, revertir). Pair with a strong English speaker; allow native-language idea generation, English on the wall sticky notes.
- **IEP/504:** Reduce SCAMPER count from 7 to 4 (S, C, M, E). Provide a pre-printed worksheet with the four prompts and example responses. Allow voice-recorded HMW.
- **Advanced learners:** After SCAMPER, run "worst possible idea" — generate 5 deliberately bad ideas, then invert each into a real one. Surface that constraints can drive creativity. Push for 5 HMWs, not 3.

Assessment

Formative: Walk during pair SCAMPER; check that pairs are pushing past obvious. Read HMWs during conferring. **Summative:** The 3 HMWs feed the Ideation Pack at Checkpoint 1 (Week 8).

Closure

Same as #6.

Homework / Extension

Capstone Journal: pick one HMW. Write 3 ideas it could lead to. Bring Wednesday.

Teacher notes

- The trap is jumping to evaluation too fast. SCAMPER is a divergent activity. Keep the wall chart for ideas only; convergence comes in Lesson 4.
- A student says "SCAMPER feels forced." Validate, then redirect: "That's the point. Forced is the technique. The forced prompt makes you think of things you wouldn't have."
- Cross-grade hook: explicitly name that Grade 9 will use the same divergent-then-convergent flow on a Business Model Canvas. The technique transfers.

Lesson 3 — Value Levers: Reading a Business Case

Grade band: 6-8 **Time:** 50 minutes **Unit:** 1 — From idea to test **Topic area:** Business analysis · BAVC strand

Objectives

By the end of this lesson, students will be able to:

- Identify which of three "value levers" (better product, lower price, better experience) a business pulled, with evidence from a 1–2 page case.
- Argue in writing which lever was most decisive in the business's success.

Standards alignment

- **CEE Standard 5 (Financial Investing), Grade 8:** Investors who own a corporation's stock or a corporate bond can earn a return — applied conceptually to "owners take risks for returns."
- **CCSS-ELA RI.8.3:** Analyze how a text makes connections among individuals, ideas, or events.
- **CCSS-ELA W.8.1:** Write arguments to support claims.
- **C3 D2.Eco.5.6-8:** Explain ways in which money facilitates exchange.

Materials

- 1–2 page case study: **In-N-Out vs. McDonald's** — provided in program assets. Anchored on a clear, kid-readable comparison: simple menu vs. broad menu, fresh-ingredient sourcing, employee pay, customer experience.
- Three-column worksheet: PRODUCT · PRICE · EXPERIENCE.
- Capstone Journal.

Vocabulary

- **Value lever** — one of three primary moves a business can make to win customers: better product, lower price, or better experience.
- **Trade-off** — when improving one lever costs you on another.

Procedure

1. Engage — Burger Test (~7 min)

Project two photos: an In-N-Out burger and a McDonald's burger. Ask: "If both were the same price, which would you pick? Why?"

Pair-share. Cold-call three students. Surface that "why" answers fall into three rough categories: taste/quality, price, experience. Name them on the board: PRODUCT · PRICE · EXPERIENCE.

2. Explain — The Three Levers (~8 min)

Define each lever with one sentence and a kid example:

- **Product** — making the actual thing better. Better fries, fresher tomatoes, a new flavor.
- **Price** — selling at a price the competition can't match (often via lower costs). $1 cheeseburger.
- **Experience** — how the customer feels around the buying. Friendly staff, fast line, clean tables, branded uniforms.

Add Cash's frame on the board: "Most businesses pull one lever harder than the others. The one they pull hardest is their bet on the customer."

Note: pulling all three hard is rare. Most businesses trade off — better product often means higher price; lower price often means weaker experience. The lever choice is a strategy.

3. Read the Case (~12 min)

Distribute the In-N-Out vs. McDonald's case. Students read silently with a highlighter, marking:

- **Yellow** — anything about the product (food itself).
- **Blue** — anything about the price (or cost structure).
- **Pink** — anything about the experience.

After reading, in pairs, complete the three-column worksheet. Each column gets at least 2 specific examples drawn from the case.

4. Apply — Defend Your Lever (~15 min)

Each student writes a 1-paragraph argument (5–7 sentences) answering: "Which value lever was most decisive for In-N-Out's success — product, price, or experience? What evidence from the case backs your call?"

Use CCSS W.8.1 frame: claim → evidence → reasoning. Confer with 4–5 students.

5. Close — Lever Spot Your Capstone (~8 min)

In the Capstone Journal:

- Which lever am I pulling hardest? (Product / Price / Experience)
- Which lever am I weakest on?
- What would have to be true for me to switch which lever I'm pulling?

Two volunteers share. End class.

Differentiation

- **ELL:** Pre-highlighted version of the case, with key evidence already marked in the three colors. Sentence frames for the argument paragraph: "I think In-N-Out pulled the **_ lever hardest because . The evidence is __**."
- **IEP/504:** Reduce case length to 1 page; provide a graphic organizer with cells already labeled. Allow oral defense of the lever instead of written.
- **Advanced learners:** Add a "trade-off" section — what did In-N-Out give up by pulling its chosen lever hardest? What's the strategic risk? Pull a second case (suggested: Costco vs. Walmart) and compare.

Assessment

Formative: Pair worksheets and conferring notes. **Summative:** Lever-spot for the Capstone feeds Unit 4's strategic-decision section in the Final PDSA Pack.

Closure

Same as #5.

Homework / Extension

Find one local business near you. Decide which lever you think it pulls hardest. Bring Wednesday with one piece of evidence.

Teacher notes

- A student says "they pull all three" — push back. "Pulling means putting effort or money into. They might do all three okay; one they invest in. Which one?"
- A student picks "experience" for a business that's clearly product-led. Coach: read the evidence again. Don't moralize — most businesses get this wrong themselves.
- Cross-grade hook: Grade 9 will name this lever choice as part of the Business Model Canvas (the value proposition box). They're learning the concept here at the level of analysis; next year they'll write it for their own venture.

Lesson 4 — Ideation Pack Studio Day (Checkpoint 1)

Grade band: 6-8 **Time:** 80 minutes (block) — Checkpoint day **Unit:** 1 — From idea to test **Topic area:** Convergent thinking · synthesis

Objectives

By the end of this lesson, students will be able to:

- Produce an Ideation Pack consisting of: a problem statement, 10 SCAMPER-driven ideas, 3 finalists with named criteria, and 1 selected with justification.

- Defend the Ideation Pack in a 3-minute peer share, citing both SCAMPER and HMW work.

Standards alignment

- **NGSS-ETS MS-ETS1-1:** Define the criteria and constraints of a design problem with sufficient precision.
- **CCSS-ELA W.8.1:** Write arguments to support claims.
- **CCSS-ELA SL.8.4:** Present claims and findings, emphasizing salient points.
- **CASEL Responsible Decision-Making.**

Materials

- Ideation Pack template (provided) — 4 pages: problem statement · 10 ideas · 3 finalists with criteria · 1 selected with justification.
- Day 1 Improvement Bet cards (returned from Lesson 1).
- Capstone Journal with SCAMPER and HMW work from Lessons 2–3.
- Wall space for the class ideation gallery.
- Rubric (Business Plan Rubric from the Rubric Bank — adapted for ideation).

Vocabulary

(All previously introduced — recycle, no new terms today.) Reinforce: SCAMPER, HMW, divergent, convergent, criteria.

Procedure

1. Engage — Hand Back Day 1 Cards (~5 min)

Hand back the Day 1 Improvement Bet cards. Quietly. Ask students to read their own and write one sentence in the Journal: "Has my Capstone topic shifted since Day 1? In what way?"

2. Page 1 — The Problem Statement (~10 min)

Mini-lesson on the problem-statement format (recycled from Grade 7):

> *"[Customer/user] struggles with [problem] when [context], because [cause]."*

Students draft on Page 1 of the Ideation Pack. Confer briefly.

3. Page 2 — The 10 Ideas (~20 min)

Students transfer their SCAMPER work + new ideas to Page 2. Format: numbered list, 1 line per idea. Goal: 10 distinct ideas.

If a student has fewer than 10 from prior work, prompt: "Apply SCAMPER again — what about Eliminate? What about Reverse? Push for 10."

Confer with each student for 2–3 minutes.

4. Page 3 — The 3 Finalists with Criteria (~20 min)

Mini-lesson on selection criteria. The class brainstorms common criteria:

- Can I actually do this with my current skills?
- Would my customer (from Grade 7 persona, if applicable) actually want it?
- Can I afford to test it (time + money)?
- Is it safe and honest?
- Could I run a test on it in 3 days?

Students pick **3 criteria** that matter for their venture, then score each of their 10 ideas against the 3 criteria using a simple 1–3 scale. The top-scoring 3 ideas are the finalists.

5. Page 4 — The 1 Selected with Justification (~15 min)

Students pick 1 of the 3 finalists and write a paragraph (5–7 sentences) explaining the choice. Required elements:

- Which idea wins.
- Why this idea over the other two finalists, in named criteria language.
- What's the first test you could run on it (preview Unit 3).

6. Gallery Walk + Peer Share (~10 min)

Tape Ideation Packs to wall. Students walk in silence for 4 minutes, leaving sticky-note feedback ("I see..." "I wonder..."). Then 3 students share their pack in 3 minutes each — 30 sec problem statement, 1 min ideation, 1 min selection, 30 sec next test.

Use rubric criteria as listening structure.

Differentiation

- **ELL:** Provide the Ideation Pack template with each section pre-labeled and a vocabulary glossary on the back. Sentence frames for the justification paragraph. Allow oral defense in pair-share rather than whole-class share.
- **IEP/504:** Pre-fill student name and date; chunk the worksheet to one section per page. Reduce idea count from 10 to 6 (the technique discipline matters more than count). Allow audio submission of the justification.
- **Advanced learners:** Add a "Plan B" section — what's the second-best idea, and what would have to happen for it to become the selected one? Connects to Grade 10's premortem work.

Assessment

Summative: The Ideation Pack is graded on the Business Plan Rubric (Rubric Bank). Use anchor papers to calibrate scoring.

Closure

Pack submitted (digital or paper) for grading.

Homework / Extension

Capstone Journal entry: "What's one thing my selected idea cannot do that I'd need to fix before launching?"

Teacher notes

- A student picks the same idea they had on Day 1 of Grade 7. That's allowed — but require evidence of the SCAMPER process (i.e., they considered alternatives and rejected them with reasoning). Sticking with last year's idea is fine; sticking without examination is not.
- A student panics during selection. Coach: "Pick the one you can run a test on by Friday. We can change it later. Don't optimize for forever — optimize for trying."
- The peer share is high-leverage; protect the time. Even if the gallery walk runs short, do all 3 share-outs.

Lesson 5 — Linear-Equation Break-Even Modeling

"Revenue is loud. Contribution margin is true."

— CASH

Grade band: 6-8 **Time:** 60 minutes (block — covers core math + application) **Unit:** 2 — The break-even truth **Topic area:** Break-even point · linear equations

Objectives

By the end of this lesson, students will be able to:

- Set up the break-even equation as a linear equation: units × price = fixed costs + variable cost × units.
- Solve for break-even volume (units) algebraically using sample numbers, with units labeled.

Standards alignment

- **CCSS-Math 8.EE.C.7:** Solve linear equations in one variable, including equations whose solutions require expanding expressions using the distributive property and collecting like terms.
- **CCSS-Math 8.F.B.4:** Construct a function to model a linear relationship between two quantities.
- **Jump$tart Standard 6 (Financial Decision Making), Benchmark 8-2:** Identify ways consumers may avoid making poor decisions. (Applied to entrepreneurs avoiding pricing mistakes.)
- **CEE Standard 1 (Earning Income), Grade 8:** People who own and run their own business take risks but also have potential for higher rewards.

Materials

- Sample business: a friendship-bracelet venture. Fixed costs: $30/month (Etsy fee + supplies station). Price per bracelet: $5. Variable cost per bracelet: $2.
- Break-even calculator template (provided) — 1-page worksheet with input cells and equation cells.
- Calculators allowed throughout.
- Capstone Journal.

Vocabulary

- **Fixed cost** — a cost that does not change with units sold (Etsy fee, app subscription).
- **Variable cost per unit** — a cost that goes up with each unit sold (recycled from Grade 7).
- **Contribution margin per unit** — price per unit minus variable cost per unit (recycled from Grade 7).
- **Break-even point** — the number of units where total revenue equals total costs.
- **Linear equation** — an equation in which the highest power of the variable is 1; graphs as a straight line.

Procedure

1. Engage — The Bracelet Setup (~7 min)

Project the sample business. Pose: "Maya runs a bracelet venture. She charges $5. Each bracelet costs her $2 in beads and cord. She also pays a $30/month Etsy fee. How many bracelets does she need to sell in a month before she stops losing money?"

Pair-share predictions; collect 3. Don't reveal yet.

2. Explain — Setting Up the Equation (~10 min)

On the board, walk through:

- **Total revenue** = price × units = $5 × n (where n is units sold).
- **Total cost** = fixed cost + variable cost × units = $30 + $2 × n.
- **Break-even** is where these are equal: $5n = $30 + $2n.

Solve algebraically:

- $5n – $2n = $30
- $3n = $30
- n = 10 bracelets.

Surface the rule: "What's actually subtracting on the left side is contribution margin per unit. Break-even = fixed cost ÷ contribution margin per unit."

Confirm: $30 ÷ $3 = 10 bracelets. Same answer, two routes.

3. We Do — Tutoring Example (~10 min)

Together, work a second example: a tutoring service. Price: $20/session. Variable cost: $2/session (printouts + small reward). Fixed cost: $40/month (subscription to a tutor-matching app).

- $20n = $40 + $2n
- $18n = $40
- n = 2.22 sessions

Round up: 3 sessions to fully break even. Confirm: contribution margin $18 × 3 = $54 ≥ fixed cost $40.

Cross-grade hook explicit: "Last year you computed contribution margin per session — $18. That number is the input to break-even. We're stacking, not starting over."

4. You Do — Break-Even Worksheet (~20 min)

Students complete the break-even calculator template for 2 sample businesses (a snack station and a custom keychain shop), then **for their own Capstone**.

Common stuck point: students with very low fixed costs ($0–$5) get small numbers and feel uncertain. Coach: small numbers are fine. The point is the math is right. Some ventures have very low break-even; that's a feature.

Common stuck point: students whose contribution margin is very low (or negative) get huge or undefined numbers. This is teaching: "You're losing money on every unit. Break-even is impossible at this price/cost combo. What needs to change?"

5. Close — Two-Sentence Summary (~7 min)

Exit ticket: "The break-even formula in one sentence is **_. The number that controls how fast you reach break-even is ___**."

(Answer: break-even = fixed cost ÷ contribution margin; the bigger the contribution margin, the faster you reach break-even.)

End class.

Differentiation

- **ELL:** Vocabulary card with the formula in plain language and Spanish cognates (equilibrio = break-even, costo fijo = fixed cost). Pair on the worksheet for the first scenario; independent for the Capstone application. Pre-translate the worksheet column headers.
- **IEP/504:** Calculator allowed throughout. Worksheet has formula already shown at top of each section. Reduce sample-business count from 2 to 1; require Capstone application.
- **Advanced learners:** Add a graph requirement — plot total revenue and total cost as two lines on the same axes. The break-even point is where they cross. Reinforces 8.F.B.4 (linear functions) directly.

Assessment

Formative: Worksheet performance and Capstone confer. **Summative:** Break-even computation feeds the Break-Even Pack at Checkpoint 2 (Week 18).

Closure

Same as #5.

Homework / Extension

Capstone Journal: "My Capstone's break-even is **_ units per month. To hit it, I'd need ___** customers per week. Is this realistic? What's my evidence?"

Teacher notes

- A student computes break-even and announces "easy, I'll just sell that many." Push back: "What's your evidence you can find that many customers? How many did your Grade 7 persona work suggest?" This braids back to customer math.
- A student's contribution margin is negative. Don't let them skip past it. "This is teaching. Either your price is too low or your variable cost is too high. Which is more changeable?"
- The graph extension is a strong follow-up for any class that includes 8th-grade math; coordinate with the math teacher if possible.

Lesson 6 — Sensitivity: What If My Numbers Are Wrong?

Grade band: 6-8 **Time:** 50 minutes **Unit:** 2 — The break-even truth **Topic area:** Sensitivity analysis · scenario thinking

Objectives

By the end of this lesson, students will be able to:

- Build a 2-scenario sensitivity table for their break-even calculator (best-case, worst-case).
- Identify which input (price, variable cost, or fixed cost) the break-even is most sensitive to.

Standards alignment

- **CCSS-Math 8.F.B.4:** Construct a function to model a linear relationship between two quantities.
- **CCSS-Math 8.SP.A.1:** Investigate patterns of association in bivariate data.
- **Jump$tart Standard 6 (Financial Decision Making), Benchmark 8-2:** Identify ways consumers may avoid making poor decisions.
- **CASEL Self-Management:** Self-monitoring under uncertainty.

Materials

- Last lesson's break-even calculators (in Capstone Journal).
- Sensitivity table template (provided): 3 columns (Worst case, Best guess, Best case) × 4 rows (Price, Variable cost, Fixed cost, Break-even units).
- Calculator.

Vocabulary

- **Sensitivity** — how much the output changes when an input changes.
- **Scenario** — a named version of the inputs (worst case, best guess, best case).
- **Most-sensitive input** — the one whose change produces the biggest output swing.

Procedure

1. Engage — A Lemonade Surprise (~7 min)

Project: a lemonade stand with break-even of 50 cups. Reveal one at a time:

- "Lemons cost 10% more this week." (Variable cost up.)
- "It rained on Saturday." (Fewer customers — units down.)
- "City charged a $20 vendor fee." (Fixed cost up.)
- "Stand next door is selling for $1 less." (Price pressure.)

After each, ask: "Does break-even go up or down?" Surface that real businesses live with shifting inputs. The number is never permanent.

2. Explain — Sensitivity Table (~8 min)

Define sensitivity. Walk through the table:

	Worst case	Best guess	Best case
Price			
Variable cost			
Fixed cost			
Break-even (units)			

Demo on the bracelet example from Lesson 5 ($5, $2, $30):

- Worst case: price $4, VC $2.50, fixed $40. CM = $1.50. BE = 40 ÷ 1.50 = 27 bracelets.
- Best guess: price $5, VC $2, fixed $30. CM = $3. BE = 10 bracelets.
- Best case: price $6, VC $1.75, fixed $25. CM = $4.25. BE = 25 ÷ 4.25 ≈ 6 bracelets.

Surface: break-even ranges from 6 to 27 — a 4×+ swing. This is sensitivity.

3. Apply — Build Your Sensitivity Table (~25 min)

Students build the table for their own Capstone. Required: at least 1 reasoning sentence per worst/best column explaining why those numbers are realistic for that scenario.

Confer with as many students as possible. Common stuck point: best-case is unrealistically rosy ("price $20, no costs!"). Coach: "What would have to be true for that price? Would your customer pay it? What evidence do you have?"

4. Identify the Most-Sensitive Input (~7 min)

Mini-discussion: which input — when it changes — moves break-even the most for your venture?

Most students will discover that contribution margin per unit (price minus variable cost) is the lever. A 10% change in price often moves break-even more than a 10% change in fixed cost.

Have students write 2 sentences in the Journal: "My most-sensitive input is **_. If it moves badly, I would ___** to protect the business."

5. Close — One Risk, Named (~3 min)

Exit ticket: "The biggest break-even risk for my venture is **_. I'd know it's hitting if ___**."

End class.

Differentiation

- **ELL:** Provide the table pre-printed with row labels and units. Sentence frames for reasoning sentences. Cognates: escenario, sensibilidad.
- **IEP/504:** Reduce to 2 columns (best guess + worst case) instead of 3. Calculator allowed. Use spreadsheet template if helpful.
- **Advanced learners:** Build a "two-variable" sensitivity — what if price AND fixed cost both move worst-case? Best-case? This previews Grade 11's cash-flow forecasting.

Assessment

Formative: Tables and conferring notes. **Summative:** Sensitivity table feeds the Break-Even Pack at Checkpoint 2.

Closure

Same as #5.

Homework / Extension

Capstone Journal: "Look up one real cost in your Capstone (a real shipping rate, a real platform fee). Replace your best-guess with the real number. Did break-even change? By how much?"

Teacher notes

- The "biggest risk" output is gold for the Final PDSA Pack — encourage students to flag it for Unit 4.
- Cross-grade hook: name explicitly that Grade 9's P&L modeling and Grade 11's cash flow forecasting are direct extensions of this — same logic, longer time horizon.
- A student says "I can't predict the future." Coach: "Right. But you can name your bet. That's what scenarios do."

Lesson 7 — Designing the 3-Day Mini A/B Test

Grade band: 6-8 **Time:** 50 minutes **Unit:** 3 — Marketing as experiment **Topic area:** Experimental design · marketing test

Objectives

By the end of this lesson, students will be able to:

- Design a 3-day A/B test of two posters (or two messages) with one variable changing, a defined success metric, and a named bias.

- Write a hypothesis sentence in the form "I think Message [A or B] will win because [reason]."

Standards alignment

- **NGSS-ETS MS-ETS1-3:** Analyze data from tests to determine similarities and differences among several design solutions.
- **CCSS-Math 8.SP.A.1:** Investigate patterns of association in bivariate data.
- **CCSS-ELA W.8.7:** Conduct short research projects.
- **CASEL Self-Management.**

Materials

- 3-day A/B test template (provided): poster-design page + tally sheet + hypothesis card.
- Two example posters from a hypothetical "study snack box" (Message A: vibes-based; Message B: specific).
- Markers, paper, tape.
- Capstone Journal.

Vocabulary

- **Hypothesis** — a specific guess about what a test will show, with a reason.
- **Control / treatment** — the version held constant vs. the version being tested.
- **Metric** — the number that decides who wins.
- **Bias** — anything that gives one side an unfair advantage in the test.
- **Lift** — the size of the difference between the two versions.

Procedure

1. Engage — Two Snack Posters (~8 min)

Project the two example posters:

- **Message A:** "The brain food smart kids actually choose."
- **Message B:** "30 grams of protein. No crash at hour two."

Ask: "Which would make you more likely to buy? Why?" Pair-share. Surface that A leans on social proof + flattery; B leans on specifics + benefit.

Tell them: "To know which actually works, what would we have to do?" Surface "test both."

2. Explain — Three Rules of A/B Testing (~10 min)

Walk through:

1. **One variable changes.** If both posters have different colors AND different words, you don't know which mattered. Hold all else equal.
2. **Define the metric before you start.** "Whichever gets more votes" is fine; whichever "feels better" isn't.
3. **Count, don't fudge.** Tally honestly. Don't count for the side you wanted.

Cross-grade hook: "Last year (Grade 7), you ran a 3-day A/B test. This year, you'll do it again — but with hypothesis, named bias, and a PDSA cycle wrapped around it."

3. Design — Build Your Two Posters (~20 min)

Students pair up (or work alone) to build the two posters for their Capstone. Rules:

- About the same product or service.
- Same length within 10%.
- Visually similar (same size, same colors, same layout — only the words change).
- Test medium named (in-school wall, classroom poster, school newspaper, paper insert).

Confer during build.

4. Write the Hypothesis + Bias (~9 min)

In the Capstone Journal, students write:

- **Hypothesis:** "I think Message **_ will win because ___**."
- **Predicted lift:** "I expect a margin of about _____ votes (or %)."
- **One bias my test cannot eliminate:** "_____."
- **Metric to count:** "_____."

Common biases to surface: poster placement (one wall has more foot traffic), poster colors, time of day, who's near the poster.

Cash's question on the wall: "Which people, and how do you know?" — here it means: which people will see this poster, and how do you know your sample is fair?

5. Close — Cross-Pair Review (~3 min)

Each student reads their two posters and hypothesis to a partner. Partner asks one of two questions: "Could the difference between A and B be more than one variable?" or "What's a bias you didn't name?"

End class. Tomorrow, Day 1 of the test.

Differentiation

- **ELL:** Provide a poster template with placement guides; messages can be drafted in native language, finalized in English. Sentence frames for hypothesis. Pair on test design.
- **IEP/504:** Reduce to one poster pair per team if doing solo would be overwhelming. Provide pre-printed templates with two text boxes. Allow voice-recorded hypothesis.
- **Advanced learners:** Push toward two messages testing two specific psychological hooks (e.g., social proof vs. concrete-benefit). Predict not just the winner but by how much (lift). Hold yourself to that prediction in the Week 26 analysis.

Assessment

Formative: Confer during poster build. **Summative:** The 3-day test feeds the Marketing Test Pack at Checkpoint 3 (Week 28).

Closure

Same as #5.

Homework / Extension

Day 2 (15-minute check-in): tally votes. Day 3: tally + start analysis.

Teacher notes

- Some students will have unequal traffic in the test environment. That's a bias — name it in the analysis.
- Cash's question explicitly: this lesson is the right place to repeat "Which people, and how do you know?" — students should feel the pull of the question on whose votes they're counting.
- A student wants to test in their personal social media account. Have a privacy conversation first (Unit 4 surfaces this fully). Default safest test: in-school, in-class, or with paper feedback.

Lesson 8 — Reading the Test: Bias, Lift, and What It Cannot Tell You

Grade band: 6-8 **Time:** 50 minutes **Unit:** 3 — Marketing as experiment **Topic area:** Test analysis · honest interpretation · first PDSA cycle

Objectives

By the end of this lesson, students will be able to:

- Compute the lift between Message A and Message B in their A/B test using counts and a percentage.
- Write a 2-sentence honest analysis: who won + what the test cannot tell.
- Set up a first PDSA cycle on the marketing message.

Standards alignment

- **CCSS-Math 8.SP.A.1:** Investigate patterns of association in bivariate data.
- **NGSS-ETS MS-ETS1-3:** Analyze data from tests.
- **CCSS-ELA W.8.1:** Write arguments to support claims.
- **CASEL Self-Management.**

Materials

- A/B test results tally sheets (from Days 1–3).
- PDSA cycle template (provided): 4 boxes — Plan / Do / Study / Act.
- Calculator.
- Capstone Journal.

Vocabulary

- **Lift** — the percentage difference between A and B (e.g., A had 25, B had 20; A's lift is 25%).
- **Inconclusive** — when the difference is small enough that you can't tell who won.
- **PDSA** — Plan, Do, Study, Act (recycled from Lesson 1, now with the Capstone).
- **Baseline** — the starting number you're comparing against.

Procedure

1. Engage — Cash's Bias Story (~7 min)

Tell or show Cash V6 ("What Could Your A/B Test Not Tell You?").

Pause after the line "More votes doesn't mean better message. It means more eyeballs."

Pose: "What's a bias you noticed in your test that you didn't predict? Quick share." Two students answer. Surface that real biases often appear after the test, not before.

2. Explain — Lift and Inconclusive (~10 min)

Walk through computing lift:

- **Step 1:** Total votes for A and B.
- **Step 2:** Lift = (winner − loser) ÷ loser × 100%.
- **Example:** A = 25 votes, B = 20 votes. Lift = (25 − 20) ÷ 20 × 100% = 25%.

Discuss "inconclusive": at small sample sizes, a 1–2 vote difference is noise, not signal. Rough rule for a 3-day classroom test: lift under 20% with under 30 total votes is inconclusive.

Tell them: "If your test is inconclusive, that's a real result. Don't make up a winner. Your next move is to design a better test, not pretend you have an answer."

3. Apply — Compute Your Lift + Honesty (~15 min)

In pairs, students compute:

- A's votes, B's votes, total.
- Lift percentage.
- Verdict: A wins / B wins / Inconclusive.
- **Two-sentence analysis:** "Message **_ won by %. My test cannot tell me whether __**."

Confer. Push back on overconfident verdicts ("A definitely won!" with 5 votes). Push back on dismissive verdicts ("Tests don't work") on tests that gave clear lift.

4. Apply — First PDSA Cycle on the Message (~15 min)

Introduce the PDSA template. Walk through with the class using a sample:

- **Plan:** "I'll change Message A's headline to be more specific. New headline: _____. Metric: votes. Target: lift over Message B by 25%."
- **Do:** "Run the new poster for 3 days, same wall, same hours."
- **Study:** "Tally and compute lift. Compare to current baseline."
- **Act:** "If lift is over 25%, adopt the new message. If under, try a third version targeting a different psychological hook."

Students fill out their own PDSA template — first cycle on their marketing message. Required: the metric is named, the target is numeric, the next test is specific.

5. Close — Honest Sentence (~3 min)

Exit ticket: "The most honest thing I can say about my A/B test is _____."

Two volunteers share. End.

Differentiation

- **ELL:** Sentence frames for the analysis and PDSA. Pre-translate "lift" and "inconclusive" with examples. Pair on lift computation.
- **IEP/504:** Calculator throughout. PDSA template with one prompt per page. Allow oral PDSA submission.
- **Advanced learners:** Push toward computing lift two ways — by raw count and by percentage of total. Compare. Add: "What's the smallest lift I would have called significant before the test? Did my actual lift exceed that bar?" (Pre-registers the hypothesis in a real-research way.)

Assessment

Formative: Lift computations and PDSA setups. **Summative:** The PDSA cycle and the honest analysis feed the Marketing Test Pack at Checkpoint 3.

Closure

Same as #5.

Homework / Extension

Run the PDSA "Do" step over the next 3 days. Bring tally to next class.

Teacher notes

- A student's lift is 200% on a 3-vote vs. 1-vote test. This is technically correct but practically meaningless. Coach: small samples don't carry information. Recommend running the test longer.
- A student's lift is 5% on a 50-vs-47 test. Coach: also inconclusive. Honest reporting matters more than dramatic numbers.
- Cash's question explicit: "Which people voted, and how do you know your sample is fair?" — repeat this. It's the year's spine question pulled into the experiment lens.

Lesson 9 — Variance: When Plan and Reality Disagree

> *"Listening is not a moment. It's a system."*
>
> — CASH

Grade band: 6-8 **Time:** 50 minutes **Unit:** 4 — Operate smarter **Topic area:** Project planning · variance analysis

Objectives

By the end of this lesson, students will be able to:

- Compute variance (actual – planned) for milestones in their 4-week project plan.
- Write a 1-page variance explanation arguing what caused the gap and what to do next.

Standards alignment

- **CCSS-ELA W.8.1:** Write arguments to support claims with clear reasons and evidence.
- **CCSS-Math 8.SP.A.1:** Investigate patterns of association in bivariate data.

- **C3 D2.Eco.5.6-8:** Explain ways in which money facilitates exchange.
- **CASEL Self-Management:** Self-monitoring.

Materials

- 4-week project plan template (filled in from Week 29 with milestones, owner, deadline).
- Actuals sheet — what actually happened in Weeks 29–30 (collected and brought to class).
- Variance template (provided): 5 rows × 4 columns (Milestone · Plan · Actual · Variance · Reason).
- Capstone Journal.

Vocabulary

- **Variance** — the difference between actual and planned (positive = ahead, negative = behind).
- **Owner** — the person responsible for a milestone.
- **Deadline** — the date by which a milestone is supposed to be met.
- **Variance explanation** — a written analysis of why a variance occurred and what to do about it.

Procedure

1. Engage — Two Plans, Two Realities (~7 min)

Project two examples:

- **Student A:** Planned to interview 5 customers in Week 29. Actually interviewed 6.
- **Student B:** Planned to interview 5 customers in Week 29. Actually interviewed 1.

Ask: "Both have variance. Which one is more interesting? Why?"

Pair-share. Surface that both are interesting — A overshot (why? was the plan too easy? did extra interviews come for free?); B undershot (what happened? scheduling? interview fatigue? wrong customer pool?).

The point: variance is data. The story is in the gap.

2. Explain — How to Read Variance (~8 min)

Walk through the variance template:

Milestone	Plan	Actual	Variance	Reason
Interview 5 customers	5	1	−4	Couldn't reach soccer team during practice week

Milestone	Plan	Actual	Variance	Reason
Build break-even calc	1 calc	1 calc	0	On time
Post 2 social posts	2	0	−2	Hadn't asked for permission

Three categories of variance to name:

- **Schedule variance** — done at a different time than planned.
- **Scope variance** — did more or less than planned.
- **Quality variance** — completed but not at planned quality.

Surface: variance is not failure. Hidden variance is the failure. The CFO equivalent rule: "Tell me bad news fast."

3. Apply — Compute Your Variance (~15 min)

Students fill out the variance template using their actual Week 29–30 data. Each row needs:

- Plan number.
- Actual number.
- Variance computed.
- One-sentence reason.

Confer. Common stuck point: a student tries to hide a missed milestone. Coach: "Variance is fine to write down. Hiding it isn't. What's the actual number?"

4. Apply — Write the 1-Page Variance Explanation (~17 min)

Students write a 1-page variance explanation in the Capstone Journal. Required structure (CCSS W.8.1):

1. **Claim:** "My biggest variance was **_, caused by ___**."
2. **Evidence:** the numbers from the variance table.
3. **Reasoning:** why this variance happened, with at least 2 contributing factors.
4. **Next move:** "For the rest of the 4-week sprint, I will _____ to address this."

Confer with as many students as possible. Sensitivity reminder: variance is a learning artifact, not a shame artifact. Coach the framing.

5. Close — One Cause, Named (~3 min)

Exit ticket: "My biggest variance came from **_. The fix in my plan is ___**."

End class.

Differentiation

- **ELL:** Sentence frames for the variance explanation. Pre-translate "variance" with cognates (Spanish: varianza, desviación). Allow oral version of the explanation.
- **IEP/504:** Variance template pre-filled with the milestone column. Calculator allowed. Reduce milestone count from 5 to 3. Allow audio submission.
- **Advanced learners:** Add a "rolling forecast" — given current variance, what's the realistic forecast for the remaining 2 weeks? Connects to Grade 11's cash-flow forecasting. Push for a separate "schedule vs. scope vs. quality" categorization.

Assessment

Formative: Variance tables and conferring. **Summative:** The 1-page variance explanation feeds the Final PDSA Pack at Checkpoint 4 (Week 36).

Closure

Same as #5.

Homework / Extension

Capstone Journal: "What's one milestone in Week 31 that I'd revise based on this variance? Write the new plan number and why."

Teacher notes

- A student's variance is zero across all milestones. This is suspicious. Coach: "Either your plan was perfect or your tracking was loose. Which is more likely?" Push them to track at finer grain.
- A student panics at a large negative variance. Validate, then coach: "Variance is a feature of operating. The story is what you do with it."
- Cross-grade hook: Grade 11's 6-week venture sprint with weekly check-ins is built directly on this — same logic, longer horizon.

Lesson 10 — Final Capstone Defense: Final PDSA Pack

Grade band: 6-8 **Time:** 90 minutes (block — or two periods) **Unit:** Capstone synthesis
Topic area: Reflection · public presentation · year-long synthesis

Objectives

By the end of this lesson, students will be able to:

- Deliver a 5-minute Capstone Defense covering: (a) the idea selected via SCAMPER + HMW, (b) the break-even and sensitivity, (c) the marketing test results, (d) the variance + PDSA cycle outcome, (e) the privacy/safety risk plan.

- Respond to two panel questions citing specific evidence from the year's Capstone.

Standards alignment

- **CCSS-ELA SL.8.4:** Present claims and findings, emphasizing salient points in a focused, coherent manner.
- **CCSS-ELA SL.8.5:** Integrate multimedia and visual displays.
- **Jump$tart Standard 6 (Financial Decision Making), Benchmark 8-2:** Identify ways consumers may avoid making poor decisions.
- **CASEL Self-Awareness + Responsible Decision-Making.**

Materials

- Each student's full Capstone Journal.
- Defense slides or poster (built in Week 35).
- Pitch Presentation Rubric (from Rubric Bank).
- Panel of 3–5 evaluators (other teachers, parents, older students, community members).
- Timer.
- Cash Video #8 (year-end summary).

Vocabulary

(All previously introduced — recycle.)

Procedure

1. Engage — The 60-Second Warm-Up (~10 min)

Pair students up. Each gives a 60-second "elevator" version of their Capstone to their partner. Partner asks one question. Repeat with a new partner. (Two rounds total — confidence warm-up before the panel.)

2. Set the Norms (~5 min)

State the day's norms clearly:

- Each student gets 5 minutes for the defense + 3 minutes for panel Q&A.
- The panel is curious, not adversarial. They're looking for evidence and reflection — not "is this a great business?"
- The audience listens silently and writes one "I noticed..." and one "I wondered..." per presenter.

Show the Pitch Presentation Rubric one more time so students know what's being scored.

Cross-grade hook: name explicitly — "The defense format you're doing today is a younger version of what Grade 11 does for an investor-style pitch and Grade 12 does for the Capstone defense. You're getting reps now."

3. Defenses (~60 min, ~12–15 students at 8 min each; longer block or split day for more)

Each student delivers the 5-minute defense with the structure:

- **The idea (1 min):** "I selected this idea via SCAMPER + HMW because [criteria]."
- **The math (1 min):** "My break-even is [N] units. My most-sensitive input is [variable]. Here's the sensitivity table."
- **The message (1 min):** "I tested two messages over 3 days. [A/B] won by [lift %]. My honest analysis is [_]. **The first PDSA cycle changed [_]**."
- **The plan + variance (1 min):** "I planned [milestones]. I delivered [actuals]. The biggest variance was [_], **caused by [_]**. The Final PDSA cycle was [____]."
- **The risk + me (1 min):** "One privacy risk and one safety risk in my venture are [_]. **My mitigation plan is [_]**. The biggest thing I learned about myself is [____]."

After each: 3 minutes of panel Q&A. Timer keeper enforced strictly.

4. Audience Reflection (~5 min)

Each audience member completes a 2-question card:

- One thing I noticed across many defenses today: __.
- One thing I'd want to learn more about: __.

Collect.

5. Final Cash Reflection (~10 min)

Play Cash Video #8 — the year-end "Operate Smarter" summary. Students write one final Capstone Journal entry:

- The biggest thing I learned this year was [____].
- The PDSA cycle taught me [____].
- I'm proud of [____].
- I'm going to keep working on [____].

6. Close (~5 min)

Hand back Day 1 Improvement Bet cards. Students re-read theirs. Two volunteers share what 36 weeks of running tests did to their original bet.

Differentiation

- **ELL:** Allow visuals to carry more weight in the defense — props, posters, charts. Sentence frames for Q&A: "I think [____] because [evidence]." Permit a same-language peer to translate panel questions on the spot if needed.
- **IEP/504:** Allow the defense to be delivered in shorter chunks if 5 min is overwhelming — 1-minute mini-segments with a 30-second pause. Permit a printed script. Audio-record version submitted in advance is also acceptable.
- **Advanced learners:** Add a "second PDSA cycle" segment — what was the next test, run after the first cycle's Act step? Push for a 1-page written addendum identifying highest-leverage next quarter.

Assessment

Summative: The defense is graded on the Pitch Presentation Rubric. Capstone Journal across the full year is a separate portfolio grade using the Reflection Writing Rubric. The Final PDSA Pack itself (variance, plan, peer feedback round, privacy/safety risk plan) is graded on the Business Plan Rubric.

Closure

Same as #6.

Homework / Extension

(End of year — none required. Optional: capstone artifact archived for Grade 9 BMC reference.)

Teacher notes

- Panel members need a 10-minute briefing before defenses start. Cover: rubric, norms, what to ask for evidence, what not to ask. Provide them with 5 sample questions: "What was the biggest surprise from your A/B test? What did your sensitivity table say about your weakest input? Walk me through your variance again. What was the hardest piece of peer feedback to hear? If you ran one more PDSA cycle, what would you test?"
- The Day 1 Improvement Bet card return at the end is the year's emotional payoff. Don't skip.
- Some students will want to skip the privacy/safety risk plan ("nothing happened"). Push back: "What's the risk that didn't happen because you mitigated it? What's the one you missed?" This is the ESL strand outcome.

Senior expert upgrade pack — Grade 8 break-even worked examples and misconceptions

Three break-even worked examples (deepening Lesson 5)

The original lesson teaches the form. Add three at three different price points.

Worked example A — Low-margin high-volume

Custom T-shirts. Price: $15. Variable cost: $10. Fixed cost: $200/month (printer rental).

- Contribution margin per shirt: $15 − $10 = $5.
- Break-even units: $200 ÷ $5 = **40 shirts/month.**

Coach: "You need to sell 40 shirts to cover fixed costs. Every shirt after that is profit. If you sell 50, profit = 10 × $5 = $50."

Worked example B — High-margin low-volume

Custom oil-painted portraits. Price: $200. Variable cost: $30. Fixed cost: $50/month (paint supplies kept on hand).

- Contribution margin: $200 − $30 = $170.
- Break-even units: $50 ÷ $170 = **0.3** — round up to **1 portrait/month.**

Coach: "The first portrait of every month covers all the fixed cost. The next ones are mostly profit. But you also need enough demand to actually sell 1 portrait — which is harder than selling 40 T-shirts."

Worked example C — Negative contribution margin (the lesson)

Lemonade. Price: $0.50. Variable cost: $0.60 (homemade fancy ingredients). Fixed cost: $20.

- Contribution margin: $0.50 − $0.60 = **−$0.10.**
- Break-even: impossible. Selling more makes the loss bigger.

Coach: "Negative contribution margin means the math doesn't work, no matter how much you sell. You either raise the price or cut the cost. Selling more is the trap."

Common Misconceptions (Grade 8)

Kid belief	What's actually true	How to coach
"If I sell more, I'll be profitable."	Only if your contribution margin is positive. With negative CM, more sales = bigger loss.	Walk through Worked Example C above. "More cups of this lemonade means more loss. Selling more is not the fix."

Kid belief	What's actually true	How to coach
"Break-even is the goal."	Break-even is the floor. The goal is profit beyond break-even. Break-even is "I stopped losing money," not "I'm winning."	"What's your real target? 2× break-even? 3×?"
"PDSA is just trying things."	PDSA has four named steps: Plan a change, Do it, Study what happened, Act on it. Each step is recorded. "Just trying things" is the failure mode PDSA prevents.	"Where's your written hypothesis? Where's the result? What did you decide?"
"A/B test results give you a winner."	Sometimes. With small samples and tied results, the test gives you direction or "run it again."	"At what sample size would you trust the result? You're at 12. Should you act, or run more?"
"Variance is a sign of failure."	Variance is data. A plan that meets reality with zero variance is rare; usually it means the plan was so loose it couldn't be wrong.	"What's your variance this week? What does it tell you about your plan?"

ROUND 2 PATCH — Common Misconceptions (Grade 8)

Common kid-belief	Reality	Coaching move
"If I sell more, Ill be profitable."	Only true if contribution margin is positive. If contribution margin is negative, every additional sale loses you money.	Run a worked example with negative contribution. Show: 100 units × −$0.50 contribution = −$50 net. Volume amplifies losses too.
"A/B test results of 8 vs. 6 means definitely the 8."	8 vs. 6 with small sample sizes is noise, not signal. Need a larger sample or a longer test.	Show a coin-flip simulation: 8 heads vs. 6 tails in 14 flips is well within random. Affirm the noise concept.
"Break-even means the business is doing well."	Break-even means zero profit. Its the floor, not the ceiling. Beyond break-even is profit; below is loss.	Walk through the linear function: revenue = price × units; cost = fixed + variable × units. Break-even is where the lines cross.
"PDSA is just trial and error."	PDSA is structured trial and error. Plan (hypothesis), Do (test), Study (measure), Act (decide). The artifacts are the difference.	Run a PDSA cycle on a real classroom problem (e.g., "we have 5 minutes of cleanup chaos at the end of class"). Document each stage.
	Variance means reality differed from plan. Variance is information,	Show a 4-week project plan with variance. Explain: "Plan was 100 units

Common kid-belief	Reality	Coaching move
"Variance means the plan was wrong."	not failure. The 1-page variance write-up explains why.	sold; actual was 70. Variance is −30 units. Why? Weather. Plan didnt account for it."

ROUND 2 PATCH — Worked Examples for Break-Even Lesson

"A little less wrong, every week."

— CASH

Cost structure: fixed costs (rent, equipment) = $200/month. Variable cost per unit = $5. Selling price = $20 per unit.

Worked Example 1 (low fixed, low volume). Break-even units = $200 / ($20 − $5) = $200 / $15 = 13.3 units → 14 units to clear break-even.

Worked Example 2 (high fixed, same price). Fixed = $1000/month. Variable = $5. Price = $20. Break-even = $1000 / $15 = 66.7 → 67 units. Same price, 5x the fixed cost → 5x the break-even volume.

Worked Example 3 (low fixed, high price). Fixed = $200. Variable = $5. Price = $40. Break-even = $200 / $35 = 5.7 → 6 units. Higher price → much lower break-even if customers will pay $40.

Anti-example. Student writes: Break-even = Fixed / Price = $200 / $20 = 10 units. Coaching: "Forgot the variable cost. Break-even uses contribution per unit (price − variable), not just price."

ROUND 2 PATCH — Multi-Class Branching Scenario (Grade 8)

"Your A/B test results are 8 vs. 6 — signal or noise?" A 2-day branching scenario.

- **Day 1.** Two posters were tested. Poster A drew 8 customers; Poster B drew 6. Students argue in pairs: signal (B is real winner) or noise (sample too small). Run a coin-flip simulation in the room — 14 flips, count heads and tails. Show that 8-6 happens randomly more than 1/3 of the time.
- **Day 2.** Design a better test — what sample size would distinguish signal from noise? (Hint: at least 30 per arm). Discuss what changed in the design.

This branching practices BAVC (value lever — better message), PSIG (SCAMPER + How Might We for the test design), MFL (light statistics on sample sizes), CBM (3-day mini A/B test), OSO (4-week project plan), LTO (paired argument), ESL (privacy of survey respondents), LRCI (PDSA: the test was Plan-Do-Study-Act).

ROUND 2 PATCH — Privacy + Safety Mini-Case (Unit 4)

Three real scenarios for the privacy/safety mini-case:

Case 1: 14-year-old sells handmade jewelry on Etsy. - Privacy risk: Etsy stores the buyers shipping address. The kid sees the address. - Safety risk: A buyer asks for a meet-up to pick up the jewelry. - 3-step mitigation each: (privacy) only ship from a P.O. box or with parent supervision; (safety) never agree to in-person meetups; route through Etsys protected platform.

Case 2: 13-year-old runs a babysitting business via word-of-mouth. - Privacy risk: Parents share the kids phone number; texts include home addresses. - Safety risk: A new family asks the kid to babysit for an overnight. - 3-step mitigation each: (privacy) use a parent-monitored phone or app like Sittercity; (safety) never overnight without a parent reference and family meeting first.

Case 3: 15-year-old sells digital art on Instagram. - Privacy risk: DMs from buyers may include phishing links or inappropriate content. - Safety risk: Buyer asks for "exclusive content" outside the platform. - 3-step mitigation each: (privacy) use a business account separate from personal; never click DM links; (safety) never move conversations off Instagram; tell a parent when uncomfortable.

ROUND 2 PATCH — Teacher Move Set (Grade 8)

- **Students A/B test result is 8 vs. 6 and they say "definitely the 8."** Say: "How big was the sample? Could 8 vs. 6 happen by chance? Lets flip a coin 14 times." Show that small differences in small samples are noise.
- **Student says "if I sell more, Ill be profitable."** Say: "Whats the contribution margin per unit? If its negative, more sales = more loss. Walk me through the math."
- **Student writes a PDSA that skips the Study phase.** Say: "What measurement told you it worked? Whats the number that changed?"
- **Student writes a variance explanation that just says "it didnt go as planned."** Say: "Why? Whats the cause? Run a 5 Whys."
- **Student wants to skip the privacy/safety lesson because "Im not online."** Say: "Most kid-run businesses touch online — even babysitting goes through phone numbers and texts. Lets walk through the closest case to your situation."

ABOUT THE HOST

Cash, in five words

curious · warm · honest · patient · specific

"I'm Cash. I don't promise wealth. I don't promise success. I promise that the moves — interview, math, message, listen — will make you a clearer thinker about money no matter what you end up doing."

— CASH

What Cash does

- **Asks questions that point to evidence.** Cash never tells; Cash asks what your evidence says.
- **Names rules sparingly.** "An assumption is something you think is true without checking. Evidence is something you observed, heard, or counted."
- **Tells short, made-up-but-realistic stories.** 30–60 seconds, shows don't tell.
- **Celebrates the move, not the win.** Revision, honesty, and evidence — not revenue.
- **Hedges deliberately.** probably, often, sometimes — never absolutes about complex topics.

What Cash never does

- Flatter ("amazing!", "great job!"), promise wealth, moralize about money.
- Reference real brands or living celebrities as endorsements.
- Surface family financial details. (Hypothetical scenarios only.)
- Pretend the world is fair. Cash is honest about systemic factors without becoming a politics class.

> *When in doubt, work one of these into the script: Which people, and how do you know? · Being wrong on Day 1 is the deal. · Revenue is loud. Contribution margin is true. · A little less wrong, every week.*

STRAND OUTCOMES

Eight Strands. One Year. Verifiable Outcomes.

Cash to Company runs the same eight strands K-12. Each year deepens the demonstration. The Grade 8 version of each strand is below.

Code	Strand	Grade 8 demonstrated outcome
BAVC	Business Awareness and Value Creation	Identify a business problem, propose a value-creation hypothesis, and revise it with customer evidence.
PSIG	Problem Solving and Idea Generation	Generate, screen, and refine ideas; document why one was chosen and the others were not.
MFL	Money and Financial Literacy	Compute revenue, variable cost, contribution margin, and breakeven for a real product or service.
CBM	Customers, Branding, Marketing	Build a persona from interview evidence; design honest messaging; A/B test two claims.
OSO	Operations, Systems, Organization	Design a simple weekly operating cadence with named roles, handoffs, and a feedback loop.
LTO	Leadership, Teamwork, Ownership	Run and document a team meeting; resolve one disagreement using shared evidence.
ESL	Ethics, Safety, Legal Awareness	Identify the line between persuasion and manipulation; flag one borderline claim and rewrite it honestly.
LRCI	Launch Readiness and Continuous Improvement	Pilot, measure, and document at least three iterations with rationale and what changed.

STANDARDS ALIGNMENT

Six Benchmarks This Year

Standards are cited at the benchmark level. Honest gaps are named when they exist. The Capstone artifact addresses each one.

Benchmark	What students must demonstrate
Jump$tart 6 (Financial Decision Making) 8-1	Identify the components of decision-making and apply them to financial decisions.
CCSS-Math 8.SP.A.1	Construct and interpret scatter plots for bivariate measurement data.
CCSS-ELA SL.8.4	Present claims and findings, emphasizing salient points in a focused, coherent manner.
C3 D2.Eco.5.6-8	Explain ways in which money facilitates exchange.
CASEL Responsible Decision-Making	Demonstrate curiosity and open-mindedness; analyze information.
CCSS-Math 7.RP.A.3	Use proportional relationships to solve multistep ratio and percent problems.

HOW TO READ THIS TABLE

Each lesson cites at least one of these benchmarks. Each Capstone checkpoint addresses two or more. The Teacher Guide includes a per-lesson standards crosswalk.

"Standards aren't a cage. They're a list of moves you're going to be glad you can make. Pick one. Practice it. Move on."

— CASH

CAPSTONE RUBRIC (4-POINT)

How We Score the Capstone

The Capstone artifact is scored on four traits, each on a 4-point scale. Honest, specific, decision-oriented work earns the high marks. Generic or assumption-laden work earns the low ones.

Trait	1 — Emerging	2 — Developing	3 — Proficient	4 — Advanced
Evidence quality	Mostly assumptions; few or no quotes.	2-3 interview quotes; partial evidence.	5+ direct quotes; varied sources; coded patterns.	Triangulated quotes plus behavioral observation; counter-evidence acknowledged.
Per-customer math	Confused costs; no contribution margin.	CM computed but with errors; one price scenario.	Correct CM and breakeven; clear pricing rationale.	CM under multiple price scenarios; LTV-lite reasoning.
Honest messaging	Vague or misleading claims.	Some specifics; some unsupported claims.	All claims supported; clean ethics check.	A/B-tested copy with rationale; resists clever-but-borderline phrasing.
Feedback system	One-time check; no documentation.	Ad-hoc feedback; partial revisions.	1-week feedback loop with 3 documented changes.	Sustained loop plus signal-vs-noise analysis plus revision log.

CALIBRATION NOTE

Score honestly. A "3" means the student met the standard the way an adult professional would. A "4" means they did something a working entrepreneur would be proud to show. Withhold "4" unless the work earns it.

SENSITIVITY AND FAMILY ENGAGEMENT

What We Don't Ask. What We Always Frame.

Money is a tool, not a virtue. Some students live in households with credit-card debt, low or no banking access, or active financial stress. The curriculum is built so that none of those students is asked to disclose, perform, or apologize for their family situation.

Rules of the room

- **Never ask about family money.** No questions about parents income, debts, accounts, or banking status.
- **Use neutral hypotheticals.** "For a sample family with a $4,000/month take-home" — not "your family."
- **Decouple wealth from worth.** Cash never says "everyone can be rich." Cash says "you will be a clearer thinker about money no matter what you end up doing."
- **Name systemic factors honestly.** Wage gaps, geography, and luck shape outcomes. The skill we build is the skill in the student hands.
- **Protect the venture.** A student Capstone is theirs. We don't require they share revenue or family contribution. The artifact is the work, not the bank account.

Family engagement

Parents get one well-framed question per quarter — never a quiz, never a money-disclosure prompt. The default frame is curiosity ("What did you notice about your customers this week?"), not interrogation ("How much did you make?").

"Money is a tool. The question is not whether your family has a lot of it. The question is what you do with the moves we are practicing — interview, math, message, listen — wherever your starting line happens to be."

— CASH

ACTIVITY INDEX

Ten Activities. One Notebook.

Every activity in this book is built to be done with a pencil, your own evidence, and the questions Cash keeps asking. Skip nothing. Date everything. Re-read in three months.

No.	Activity	What you will do
01	Day 1 Customer Assumption	One sentence, one card. Don't try to be right; try to be specific.
02	Open vs. Leading Questions	Rewrite five leading questions into open ones. Practice on a partner.
03	Coding Patterns from Transcripts	Highlight surprises, frustrations, wishes. Tally them. Find the pattern.
04	Persona Pack v1	Combine evidence into one specific person — name, behaviors, top frustration.
05	Variable Cost per Unit	From a real receipt, compute the cost per one of the thing you sell.
06	Contribution Margin Worksheet	Price minus variable cost. Decide if it covers your fixed costs.
07	Two-Message A/B Test	Two honest messages. Three-day test. Report the winner and why.
08	Persuasion vs. Manipulation Line	Sort 10 ad claims onto a line. Defend your placement.
09	1-Week Feedback Loop	Collect, code, and act on feedback. Document three changes.
10	Final Capstone Defense	Five slides. Five minutes. Show the moves you made.

NOTEBOOK SETUP

Get a single notebook for the whole year. Number every page. The first 5 pages are your Vocabulary Spine. The last 5 pages are your Day-1 Assumptions (you will come back to those). Everything in between is your work.

Notebook checklist

- **Title page** — your name, your venture working-title, today's date.
- **Vocabulary spine** — 5 pages reserved for definitions you will add as we go.
-

Customer notes — one section for interview transcripts. Use direct quotes, with initials and dates.

- **Math section** — receipts taped in, per-unit computations, contribution margin scratch work.
- **Messaging section** — drafts of two-claim A/B tests, with what won and why.
- **Feedback log** — one page per week of your feedback system, with the three changes you made.

WORKED EXAMPLE

Worked Example: Measure What Matters

Priya runs a tutoring micro-business helping 6th graders with pre-algebra. She has eight regular clients. Five say they would recommend her. Three would not. Her old metric was "hours billed." Her new metric: "would recommend."

The new metric tells a different story. Two of her three non-recommenders share the same complaint: "Sessions feel like quizzes." Priya redesigns the first 10 minutes of every session to start with what the student is stuck on, not what Priya wants to teach. Three weeks later: 7 of 8 would recommend.

What changed? Not effort. Not skill. The metric. "Would recommend" surfaced the experience. Hours billed surfaced nothing.

WHAT TO COPY

The move here is not the answer. It is the structure: specific number, specific person, specific decision. Use that pattern in your own venture this quarter.

> *"Most adults skip this. You won't. Pick one customer. Compute one number. Decide one thing." — Cash*

END-OF-QUARTER REFLECTION

Four Quarters. Same Five Questions.

At the end of every quarter you will answer the same five questions. The answers should look different over time. Reread the previous quarter before you write a new one.

1. **Which people did you talk to this quarter, and what did they actually say?**

   Specific quotes, initials, dates. Resist the urge to summarize.
2. **What did you change in your venture because of evidence?**

   Name the change. Name the evidence that caused it.
3. **What is one number you know better now than three months ago?**

   Variable cost per unit. Contribution margin. Anything quantitative.
4. **What is a claim you made that you would take back?**

   Be honest. The claim does not have to be public. The honesty is the move.
5. **What would Cash ask you about all of this?**

   "Which people, and how do you know?" is always available. Find your own version.

"Reflection is not a feeling check-in. It is a memory tool. The you in three months wants to know what the you today learned. Write it down so future-you can use it."

— CASH

One promise.

Cash to Company makes one promise: students who move through this program will be clearer thinkers about money and entrepreneurship — no matter what they end up doing for a living.

We do not promise wealth. We do not promise success. We promise clarity.

"A little less wrong, every week."

— CASH, GRADE 8